

Unlimited Theorems

Abstract

Theorems and results have been organized into sections and subsections. Definitions, if needed, are provided at the beginning of each section and subsection. A subsection has final index 0 if and only if it is a subsection of definitions. For example, subsections 5.0, 3.3.0, and 0.0 are definition subsections, but 2.3 and 1.4.1 are not.

If a term is defined more than once in this document, the definition provided at most specific layer will apply. For example, if section 5.0 defines “ring” to not necessarily be commutative, but 5.3.0 declares a “ring” to be commutative with respect to multiplication, then in all theorems 5.3.X, “ring” will mean commutative ring, unless further overridden. Mentions of rings in 5.2, on the other hand, will not imply commutativity unless the definition is overridden.

If a term is used but not defined within a section or subsection, the definition that is “outer-most” applies. For example, if the term “homomorphism” is encountered in section 4.1, but is defined at 2.1.0, 3.0, and 5.2.3.0, then the definition in 3.0 applies. If there is no single “outer-most” definition, then the earliest of the tied definitions applies. For example, if the same term is defined in 2.0, 3.0, and 4.1.0, then its use in section 5 will correspond to the definition in 2.0.

An appendix is provided at the end of the document listing every term, along with references to each of its definitions, in alphabetical order.

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1 Logic

1.0 Definitions

Basic Set Theory

Definition 1.0.1 (Set). A *set* is an unordered collection of items; the items are usually called elements.

Definition 1.0.2 (Subset). Let X be a set. A set A which satisfies $x \in A \implies x \in X$ is called a *subset* of X . We write $A \subset X$ to mean A is a subset of X .

Definition 1.0.3 (Set Equality). Two sets A, B are *equal* if $A \subset B$ and $B \subset A$, denoted $A = B$.

Definition 1.0.4 (Power Set). Let X be a set. The *power set* of X , denoted $\mathcal{P}(X)$ is defined to be the set of all subsets of X .

Definition 1.0.5 (Empty Set). The set with no elements is called the *empty set*, usually denoted ϕ .

Definition 1.0.6 (Ordered List). An *ordered list* is a collection of items specified by an order, with possible repetition. An ordered list of n elements can be called an n -tuple. An ordered list of 2 elements is sometimes called a pair.

Definition 1.0.7 (Cartesian Product). Let A, B be sets. The *Cartesian product*, or simply *product*, of A and B is the set $A \times B := \{(a, b) : a \in A, b \in B\}$.

Definition 1.0.8 (Union). Let A, B be sets. The *union* of A and B is the set $A \cup B := \{x : x \in A \text{ or } x \in B\}$.

Definition 1.0.9 (Intersection). Let A, B be sets. The *intersection* of A and B is the set $A \cap B := \{x : x \in A \text{ and } x \in B\}$.

Definition 1.0.10 (Set Difference). Let A be a set, and $B \subset A$ a subset. The *set difference* is the set of elements in A but not B , denoted $A \setminus B := \{a \in A : a \notin B\}$.

Definition 1.0.11 (Complement). Let $B \subset A$ be a subset. The set $B^c = A \setminus B$ is called the *complement* of B in A .

Definition 1.0.12 (Disjoint). Let A, B be sets. A and B are called *disjoint* if $A \cap B = \phi$.

Definition 1.0.13 (Cover). Let X be a set, and $B = \{U_i\}$ be a collection of sets. B is called a *cover* of X if $X \subset \bigcup_i U_i$. A subset $C \subset B$ that is also a cover is called a *subcover*.

Definition 1.0.14 (Relation). Let A be a collection. A *relation* R on A is a subset $R \subset A \times A$. For each $x, y \in A$, we denote $xRy \iff (x, y) \in R$.

Definition 1.0.15 (Transitive Relation). Let R be a relation on a collection A . R is called *transitive* if, for each $x, y, z \in A$, if xRy and yRz , then xRz .

Definition 1.0.16 (Symmetric Relation). Let R be a relation on a collection A . R is called *symmetric* if, for each $x, y \in A$, if xRy , then yRx .

Definition 1.0.17 (Reflexive Relation). Let R be a relation on a collection A . R is called *reflexive* if, for each $x \in A$, if xRx .

Definition 1.0.18 (Equivalence Relation). Let R be a relation on some collection. R is called an *equivalence relation* if it is transitive, symmetric, and reflexive.

Definition 1.0.19 (Equivalence Class). Let R be an equivalence relation on a collection A , and $a \in A$. The set $[a]_R = \{x \in A : xRa\}$ is called the *equivalence class* of a . We sometimes simply write $[a]$ if the relation is understood.

Definition 1.0.20 (Function, Map). Let A, B be non-empty sets. A *function* or *map* f from A to B is a rule that assigns to each element $a \in A$ a unique element $b \in B$. A is called the *domain* of the function, and B the *codomain*. The signature of the function is typically denoted $f : A \rightarrow B$.

Definition 1.0.21 (Injection). Let $f : A \rightarrow B$ be a map. f is an *injection*, and said to be *injective* if, for any $x, y \in A$, $f(x) = f(y) \implies x = y$.

Definition 1.0.22 (Surjection). Let $f : A \rightarrow B$ be a map. f is a *surjection*, and said to be *surjective* if, for every $y \in B$, there is some $x \in X$ so that $y = f(x)$.

Definition 1.0.23 (Bijection). Let $f : A \rightarrow B$ be a map. f is a *bijection*, and said to be *bijective* if it is injective and surjective.

Definition 1.0.24 (Inverse). Let $f : A \rightarrow B$ be a bijection. The map $f^{-1} : B \rightarrow A$ is called the *inverse* of f is, for each $a \in A, b \in B$, $f(f^{-1}(b)) = b$ and $f^{-1}(f(a)) = a$.

Definition 1.0.25 (Operation). Let A be a set. A function f with signature $f : A \times A \rightarrow A$ is called an *operation* or *operator* on A . The evaluation $f(x, y)$ is often denoted xy , or simply xy .

Definition 1.0.26 (Image). Let $f : A \rightarrow B$ be a function. The set $\text{Im} f := \{f(x) : x \in X\}$ is called the *image* of f .

Definition 1.0.27 (Pre-Image). Let $f : A \rightarrow B$ be a function. Let $C \subset B$ be a subset. The set $f^{-1}(C) := \{x \in A : f(x) \in C\}$ is called the *pre-image* of C .

Definition 1.0.28 (Cardinality). Let U be the collection of all sets. Define the relation $\sim$ on U so that $A \sim B \iff \exists f : A \rightarrow B$ so that f is bijective. The *cardinality* of a set X is the equivalence class $[X]_{\sim}$. The cardinality of ϕ is denoted as 0.

Definition 1.0.29 (Singleton). Let A be a set. A is called a *singleton* if it has the same cardinality as the set $\{\phi\}$. The cardinality of a singleton is denoted as 1.

Definition 1.0.30 (Natural Numbers). Define a map S which sends a set X to the set $X \cup \{X\}$. We say a set U is a **successor** of X if and only if it is either $S(X)$ or the successor of a successor of X . The **natural numbers** are given by the set $\mathbb{N} = \{\text{Cardinalities of successors of } \phi\}$. The chain of subsets $S(\phi) \subset S(S(\phi)) \subset S(S(S(\phi))) \dots$ can be denoted the list $1, 2, 3 \dots$.

Definition 1.0.31 (Finite). Let A be a set. A is called **finite** if it is empty or a member of some $n \in \mathbb{N}$.

Definition 1.0.32 (Infinite). Let A be a set. A is called **infinite** if it is not finite.

1.1 Set-Map Compatibility

Theorem 1.1.1. Let $f : A \rightarrow B$, $X \subset A$. Then, $f(X^c) = f(X)^c$.

2 Topology

2.0 Definitions

Definition 2.0.1 (Topology). Let X be a set. A subset τ of the power set $\mathcal{P}(X)$ is called a *topology* on X if the following are true:

- $\phi, X \in \tau$
- Any arbitrary union of elements of τ belongs to τ
- Any finite intersection of elements of τ belongs to τ .

Definition 2.0.2 (Topological Space). Let X be a set, and τ a topology on X . The pair (X, τ) is called a *topological space*.

Definition 2.0.3 (Open Set). Let (X, τ) be a topological space. A subset $U \subset X$ is called *open* in X if $U \in \tau$.

Definition 2.0.4 (Closed Set). Let (X, τ) be a topological space. A subset $U \subset X$ is called *open* in X if the set $X \setminus U$ is open.

Definition 2.0.5 (Neighborhood). Let X be a topological space, and $x \in X$ a point. A subset $N \subset X$ is called a *neighborhood* of x if there is some subset $U \subset N$ so that U is open and $x \in U$.

Definition 2.0.6 (Subspace Topology). Let (X, τ_1) be a topological space, and $Y \subset X$. The *subspace topology* on Y is the set $\tau_2 = \{U \cap Y : U \in \tau_1\}$. The pair (Y, τ_2) is called a *subspace* of X .

Definition 2.0.7 (Open Cover). Let A be a topological space. A cover B of A is called an *open cover* if every set in it is open.

Definition 2.0.8 (Continuous Map). Let $(X, \tau_1), (Y, \tau_2)$ be topological spaces. A map $f : X \rightarrow Y$ is called *continuous* if the pre-image of any open set in Y is an open set of X .

Definition 2.0.9 (Homeomorphism). Let $f : X \rightarrow Y$ be a continuous bijection. Then, f is called a *homeomorphism* if its inverse is continuous.

Definition 2.0.10 (Compact Space). Let $A \subset X$ be a subspace. A is called *compact* if for every open cover B of A , there is an open subcover $C \subset B$ such that C is finite.

Definition 2.0.11 (Hausdorff Space). Let X be a topological space. X is called *hausdorff* if, for any $x, y \in X$, if $x \neq y$, then there exist neighborhoods U of x and V of y so that U and V are disjoint.

Definition 2.0.12 (Homotopy). Let $f, g : X \rightarrow Y$ be continuous functions. A *homotopy* between f and g is a continuous map $H : X \times [0, 1] \rightarrow Y$ satisfying $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$.

2.1 Point Set Topology

Theorem 2.1.1. Let $f : A \rightarrow B$ be a map between topological spaces. The following are equivalent:

- f is continuous.
- $K \subset B$ is closed if and only if $f^{-1}(K)$ is closed.

Theorem 2.1.2. Let X be a topological space. The following are equivalent:

- $U \subset X$ is an open set.
- For each $a \in U$, there is some $K \subset U$ so that $a \in K$ and K is open.
- U is a neighborhood of all of its points.

Theorem 2.1.3. Let X be compact and Y be hausdorff. A continuous bijection $f : X \rightarrow Y$ is a homeomorphism.

Proof. It suffices to show that f^{-1} is continuous. Let $A \subset X$ be an open set. Let $y \in f(A)$.

3 Abstract Algebra

3.0 Definitions

Definition 3.0.1. Let G be a set. Let $\circ$ be an operation on G . The pair $(G, \circ)$ is called a *group* if the following are true:

- There exists an element $e \in G$ so that $e \circ g = g \circ e = g$ for any $g \in G$, called the *identity*
- For any $g \in G$, there exists an *inverse* $g^{-1} \in G$ so that $g \circ g^{-1} = g^{-1} \circ g = e$
- For any $a, b, c \in G$, $(a \circ b) \circ c = a \circ (b \circ c)$.

The operation $\circ$ is often omitted.

4 Representation Theory

4.0 Definitions

Appendix

A Definition Index

B Notation